

Operator Splitting in the Dual

Forward - Backward: $\underbrace{(\mathbf{I} + \lambda \partial f)^{-1} (\mathbf{I} - \lambda \nabla g)}$

proximal - gradient algorithm.

[Now will apply DR: $\frac{1}{2}(\mathbf{I} + \mathbf{G} \circ \mathbf{f})$]
in the dual.

$$\mathbf{C} = 2\mathbf{R} - \mathbf{I}$$

$$\mathbf{R} = (\mathbf{I} + \lambda \partial f)^{-1}$$

Recall our Canonical Problem

$$\min: f_1(x_1) + f_2(x_2)$$

$$\text{s.t.: } A_1 x_1 + A_2 x_2 = b$$

$$\max: -f_1^*(-A_1^T z) - f_2^*(-A_2^T z) - b^T z$$

Recall: • If both f_1, f_2 are strongly convex

⇒ use dual grad ascent: separable Lagrangian

• If only f_1 is strongly convex?

⇒ dual prox grad: separable into Lagrangian + Aug. Lag.

• If neither f_1 nor f_2 str. convex: PPA in dual

⇒ Aug. Lagrangian — coupled.

DR in the Dual

$$\begin{aligned} \max: & \underbrace{f_1^*(-A_1^T z)} + \underbrace{f_2^*(-A_2^T z)} + \underbrace{b^T z} \\ \min: & \underbrace{\qquad\qquad\qquad}_g \end{aligned}$$

① $u_{k+1} = \text{prox}_{f_1}(z_k + w_k)$

② $z_{k+1} = \text{prox}_{f_2}(u_{k+1} - w_k)$

③ $w_{k+1} = w_k + z_{k+1} - u_{k+1}$

④ $u_{k+1} = \text{prox}_{f_1}(z_k + w_k), \quad g = f_1^*(-A_1^T z) + b^T z$

⑤ $\hat{x}_1 = \arg\min_{x_1} : f_1(x_1) + (z_k + w_k)^T (A_1 x_1 - b) + \frac{\lambda}{2} \|A_1 x_1 - b\|_2^2$

$u_{k+1} = \underbrace{(z_k + w_k)} + \underbrace{\lambda (A_1 \hat{x}_1 - b)}$

⑥ $\hat{x}_2 = \arg\min_{x_2} : f_2(x_2) + z_k^T (A_2 x_2) + \frac{\lambda}{2} \|A_1 \hat{x}_1 + A_2 x_2 - b\|_2^2$

$z_{k+1} = z_k + \lambda (A_1 \hat{x}_1 + A_2 \hat{x}_2 - b)$

⑦ $w_{k+1} = \lambda A_2 \hat{x}_2$

Exercise:

Check all details

ADMM - Alternating Direction Method of Multipliers

$$\begin{aligned}\hat{x}_1 &= \underset{x_1}{\operatorname{argmin}}: f_1(x_1) + (z+w)^T (A_1 x_1 - b) + \frac{\lambda}{2} \|A_1 x_1 - b\|_2^2 \\ (?) &= \underset{x_1}{\operatorname{argmin}}: f_1(x_1) + z^T (A_1 x_1 - b) + \frac{\lambda}{2} \|A_1 x_1 + A_2 x_2 - b\|_2^2\end{aligned}$$

$$\hat{x}_2 = \underset{x_2}{\operatorname{argmin}}: f_2(x_2) + z^T A_2 x_2 + \frac{\lambda}{2} \|A_1 \hat{x}_1 + A_2 x_2 - b\|_2^2$$

$$z^+ = z + \lambda (A_1 \hat{x}_1 + A_2 \hat{x}_2 - b)$$

Note: $L(x_1, x_2, z) = f_1(x_1) + f_2(x_2) + z^T (A_1 x_1 + A_2 x_2 - b) + \frac{\lambda}{2} \|A_1 x_1 + A_2 x_2 - b\|_2^2$

$$\left[\begin{array}{l} \hat{x}_1 = \underset{x_1}{\operatorname{argmin}}: L(x_1, x_2, z) \\ \hat{x}_2 = \underset{x_2}{\operatorname{argmin}}: L(\hat{x}_1, x_2, z) \\ z^+ = z + \lambda (A_1 \hat{x}_1 + A_2 \hat{x}_2 - b) \end{array} \right]$$

Comparisons to Other Dual Methods

Exercise:

Compare to:

- Grad Ascent
- Prox Pt Alg
- Prox Grad

all in dual.

Extra Splitting Variables (1/2)

$$\min; f_1(x) + f_2(Ax) \quad \swarrow \boxed{\text{prox}_{f_1}, \text{prox}_{f_2} \text{ "easy"}}$$

$$\min; f_1(x_1) + f_2(x_2)$$

$$\text{st: } Ax_1 = x_2$$

Exercise : ADMM

?

easy

$$\hat{x}_1 = \arg \min_{x_1} f_1(x_1) + f_2(x_2) + \frac{\lambda}{2} \|Ax_1 - x_2 + z/\lambda\|_2^2$$

$$\hat{x}_2 = \arg \min_{x_2} f_1(\hat{x}_1) + f_2(x_2) + \frac{\lambda}{2} \|A\hat{x}_1 - x_2 + z/\lambda\|_2^2$$

$$z_{k+1} = \text{---}$$

why might this be hard?

Extra Splitting Variables (2/2)

$$\min: f_1(x_1) + f_2(x_2)$$

$$\text{st: } \begin{bmatrix} A \\ 1 \end{bmatrix} x_1 = \begin{bmatrix} x_2 \\ x_3 \end{bmatrix}$$

Now: ADMM requires the evaluation

of $\text{prox}_{\frac{1}{\lambda} f_1}$, $\text{prox}_{\frac{1}{\lambda} f_2}$

$$\hat{x}_1 = \arg \min_{x_1} f_1(x_1) + f_2(x_2) + \frac{\lambda}{2} \|Ax_1 - x_2\|_2^2 + \frac{\lambda}{2} \|Ax_1 - x_3 + \frac{2}{\lambda}\|_2^2$$

$$(\hat{x}_1, \hat{x}_2) = \arg \min_{x_1, x_2} \quad \text{same} \quad \uparrow \quad \uparrow \quad \uparrow$$

Easier than previous slide.